

Shujah Ur Rehman

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RESEARCH INTERESTS

Computer Vision, Deep Learning for Robotic Perception, Medical Image Segmentation, Multi-modal (RGB-D) Sensor Fusion, Semi-Supervised & Self-Supervised Learning

EDUCATION

- **Lahore University of Management Sciences (LUMS), Pakistan** Aug 2021 – Jan 2024
M.S. Electrical Engineering, Graduated with Distinction
- **Lahore University of Management Sciences (LUMS), Pakistan** Aug 2015 – Jul 2019
B.S. Electrical Engineering

RESEARCH EXPERIENCE

- **Research Specialist, RISC Lab, KAUST** Sep 2025 – Present
Perception for Agricultural and Underwater Robotics
Saudi Arabia
 - Designing and deploying an end-to-end computer vision pipeline for autonomous date-palm harvesting, running on Jetson AGX Orin at ~30 FPS via INT8/FP16 quantization and model scaling.
 - Curating and annotating a 7,000+ image dataset using CLIP-based filtering and instance segmentation for challenging outdoor agricultural conditions.
 - Training and optimizing YOLO-based segmentation models for real-time fruit detection and ripeness analysis; integrating perception outputs with ROS2 robotic manipulation pipelines.
- **Research Associate, ATP Lab, LUMS** Sep 2022 – Aug 2025
Semi-Supervised Learning for Medical Image Segmentation
Pakistan
 - Designed a computationally efficient dual-decoder U-Net for 3D cardiac MRI segmentation in PyTorch, achieving state-of-the-art performance on public benchmarks.
 - Built complete pipelines spanning preprocessing, semi-supervised training, and evaluation for 2D and 3D cardiac MRI datasets.
 - Applied self-supervised pretraining and fine-tuning strategies to improve model generalizability across multi-center medical datasets.
- **Research Assistant, Computer Vision & Graphics Lab, LUMS** Jul 2022 – Oct 2022
Crop Classification for the LBDC Region, Punjab
Pakistan
 - Collaborated with a research team to improve the classification accuracy of a Random Forest classifier for Rabi and Kharif seasons using Google Earth Engine.
 - Developed spectral index visualizations (NDVI, EVI, NDWI) and conducted quantitative analysis on thresholding and masking strategies to improve classification accuracy.
- **Research Officer, HPCNL, UET Lahore** Mar 2021 – Aug 2021
Cross-Lingual Transfer Learning for Urdu Text
Pakistan
 - Pretrained a BERT-based transformer model from scratch on an Urdu corpus, covering the full pipeline from raw text preprocessing to model training.
 - Built NLP pipelines for low-resource Urdu text tasks (propaganda detection, topic classification, sentiment analysis) using cross-lingual transfer learning; benchmarked transfer learning methods and published findings at ICOSST 2021.

TEACHING EXPERIENCE

- **CS 437: Deep Learning, LUMS** Sep 2022 – Dec 2023
Teaching Assistant (two semesters)
 - Developed assignments and tutorial materials; held weekly office hours, invigilated exams, and graded coursework.
- **CS 653: Digital Image Processing, LUMS** Jan 2023 – May 2023
Teaching Assistant
 - Held weekly office hours, invigilated quizzes, and graded assignments.

PUBLICATIONS

- **Investigating Cross-Lingual Transfer Learning for Urdu Text Using Word Embeddings** Dec 2021
International Conference on Open-Source Systems and Technologies (ICOSST) 2021

TECHNICAL SKILLS

- **Languages:** Python, C++, MATLAB, Bash
ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers
Computer Vision: OpenCV, YOLO (v5/v8), Instance Segmentation, CLIP, INT8/FP16 Quantization
Robotics & Deployment: ROS2, Jetson AGX Orin (edge inference), Docker
Tools: Google Earth Engine, NumPy, Pandas, Matplotlib, Seaborn, L^AT_EX